

UNSUPERVISED MACHINE LEARNING

Patient Segmentation with K-Means Clustering

*Identifying which hospital patients need proactive care, using unsupervised learning on
behavioral and clinical data*

Prepared by: [Aleeza Nadeem]

Dataset: Hospital Data (Kaggle) — Patient Profiling Study

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1. Executive Summary

This project applies K-Means clustering — an unsupervised machine learning technique — to segment hospital patients into distinct, actionable groups using behavioral and clinical data (visit frequency, stay duration, spending, BMI, and chronic condition indicators). No pre-existing labels were used; the goal was to let the algorithm discover natural patient groupings directly from the data.

Three patient segments were identified: a Low-Risk / Healthy group requiring minimal intervention, a High-Complexity / Inpatient group best suited for proactive care coordination, and a Frequent Monitoring / High-BMI group requiring consistent outpatient attention despite a healthier profile on paper. These segments give care teams a concrete way to prioritize limited resources rather than treating every patient identically.

2. Business Problem

Hospitals routinely treat large volumes of patients without a systematic way to distinguish those who need proactive attention from those who don't. Treating every patient the same wastes resources on low-need cases while leaving genuinely high-complexity patients under-monitored until a costly, avoidable readmission occurs.

Published healthcare cost research consistently shows a small fraction of patients accounts for a disproportionate share of total spend — often driven by frequent readmissions or ongoing complex care needs. This project's objective was to surface that high-need group, and other meaningful segments, directly from patient data — without relying on a clinician to manually review every record.

3. Dataset Overview

Source: "Hospital data" (Kaggle, by ibnarahat) — a synthetic patient profiling dataset covering demographic, clinical, and utilization attributes.

Column	Description
Patient_ID	Unique identifier (excluded from modeling)
Age	Patient age
Gender	Male / Female
Blood_Type	Blood type
Chronic_Condition	0–4 scale — severity or count of chronic conditions
BMI	Body Mass Index
Annual_Visits	Number of hospital visits per year
Avg_Stay_Duration	Average length of stay, in days

Column	Description
Total_Spending	Total annual cost
Insurance_Type	Insurance category; substantial missing values likely represent uninsured patients rather than missing data

4. Methodology

4.1 Exploratory Data Analysis

- Checked for duplicate records and missing values across all columns.
- Identified meaningful missingness in Insurance_Type; treated as its own “None” category rather than imputed, since it likely reflects uninsured status.
- Reviewed skew and outliers on numeric columns to determine whether transformation was warranted.

4.2 Feature Engineering & Selection

- Dropped Patient_ID — an identifier, not a behavioral signal.
- Selected Age, Chronic_Condition, BMI, Annual_Visits, Avg_Stay_Duration, and Total_Spending as clustering features, since these directly capture patient utilization and clinical complexity.
- Reserved Gender and Insurance_Type for post-hoc cluster profiling rather than using them to drive the clustering itself.

4.3 Preprocessing

StandardScaler was applied to all clustering features (mean = 0, std = 1) so that no single feature — particularly Total_Spending, which has the largest raw numeric range — dominates the Euclidean distance calculations that K-Means relies on.

4.4 Choosing K

The Elbow Method and Silhouette Score were computed across K = 2 through 7 to evaluate candidate cluster counts objectively rather than by guesswork.

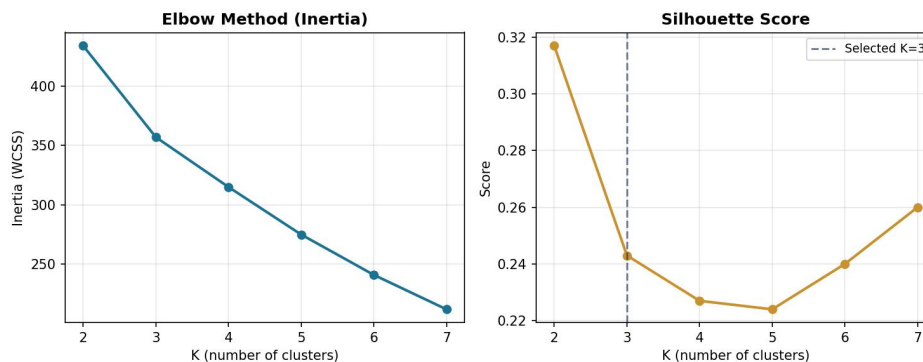


Figure 1. Elbow Method (left) and Silhouette Score (right) across K = 2–7.

The Elbow curve declined gradually with no sharp bend, offering weak standalone evidence. The Silhouette Score peaked at K=2 (0.317), with K=3 close behind (0.243).

K=3 was selected over the statistically highest-scoring K=2 because it revealed a third, genuinely distinct patient persona — frequent-visit, high-BMI outpatients — that K=2 was folding into a broader, less actionable “low-need” group. This is a case where interpretability and business usefulness outweighed a marginally higher Silhouette score, and it was validated by comparing the detailed cluster profiles at both K values before deciding.

4.5 Model & Validation

Final model: `KMeans(n_clusters=3, init='k-means++', n_init=10, random_state=42)` — k-means++ initialization and 10 independent runs were used to reduce the risk of a poor local minimum from unlucky random starts.

Principal Component Analysis (PCA) was used to compress all six clustering features into two dimensions for visual validation of cluster separation, since the model uses more than two features and cannot be visualized directly.

5. Results

5.1 Cluster Profiles (K = 3)

Cluster	Age	Chronic Cond.	BMI	Annual Visits	Avg Stay (days)	Total Spending	Count
0 — Low-Risk / Healthy	37.1	1.1	33.8	2.2	2.4	\$924.5	38
1 — High-Complexity / Inpatient	60.0	1.5	39.1	4.1	6.0	\$1,794.1	36
2 — Frequent Monitoring / High-BMI	38.1	1.5	44.8	5.9	2.6	\$1,262.3	35

5.2 Persona Summaries

- Low-Risk / Healthy — younger, low visit frequency, short or no stays, lowest spending. Minimal intervention needed; standard preventive care is sufficient.
- High-Complexity / Inpatient — older, longest average stays, highest spending. Best candidates for proactive care coordination and readmission-prevention programs — the group most likely to drive disproportionate cost if left unmanaged.
- Frequent Monitoring / High-BMI — similar age to the healthy group but the highest visit frequency and BMI, despite short individual stays. Suggests an actively-managed, weight-related condition requiring frequent outpatient monitoring rather than inpatient care — a distinct care pathway from Cluster 1.

5.3 Cluster Separation (PCA View)

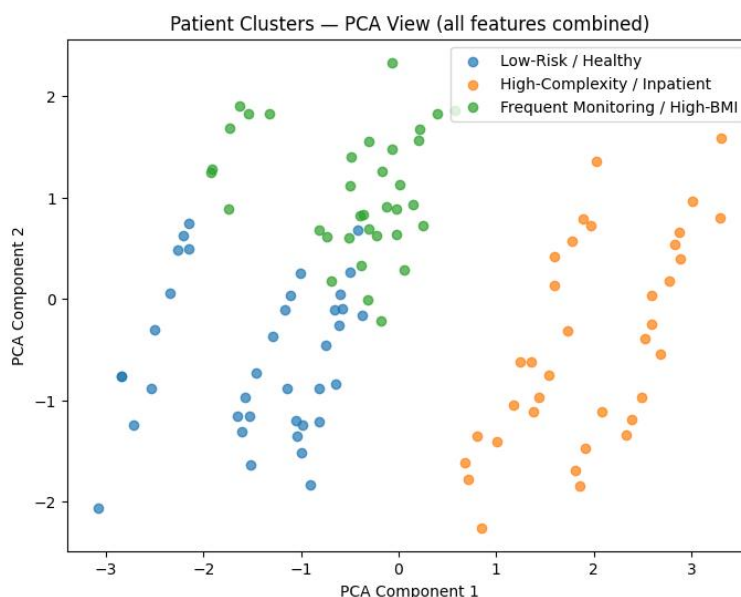


Figure 2. All six clustering features compressed into two dimensions via PCA. High-Complexity and Frequent-Monitoring clusters are clearly separated; the Low-Risk cluster shows natural, expected overlap at its edges.

6. Business Recommendations

- Route Cluster 1 (High-Complexity / Inpatient) patients into proactive care coordination — this group carries the highest cost and longest stays and is the highest-value target for early intervention.
- Apply targeted outpatient monitoring protocols to Cluster 2 (Frequent Monitoring / High-BMI) patients — their age and stay length alone would suggest low risk, but visit frequency and BMI indicate an ongoing condition needing consistent attention.
- Maintain standard care pathways for Cluster 0 (Low-Risk / Healthy) patients, and reallocate resources saved here toward Clusters 1 and 2 — this reallocation is the core efficiency gain this segmentation enables.
- Treat statistically “best” K and “most actionable” K as separate questions — validate any cluster count against domain interpretability, not the evaluation metric alone.

7. Limitations

- Cluster boundaries reflect only the six features selected; incorporating lab results, diagnosis codes, or medication history could reveal further sub-segments.
- The exact meaning of Chronic_Condition (severity scale vs. condition count) was inferred rather than confirmed against the original data source, and should be verified before any clinical use.
- K-Means assumes roughly spherical, similarly-sized clusters. The moderate Silhouette scores observed (≈ 0.24 – 0.32) indicate patient behavior here is more continuous than cleanly separated — an expected characteristic of real-world health data, stated explicitly rather than overstating cluster purity.

8. Tools & Technologies

pandas · NumPy · scikit-learn (KMeans, StandardScaler, PCA, silhouette_score) · matplotlib

9. Next Steps

- Compare against DBSCAN to test whether density-based clustering finds the same three groups or reveals additional structure and outliers K-Means cannot detect.
- Build an interactive interface (e.g., Streamlit) so care teams can classify a new patient's segment at intake, mirroring the approach used in the companion customer-segmentation project.